

3.1 Practice - Solve and Graph Inequalities

Draw a graph for each inequality and give interval notation.

1) $n > -5$

2) $n > 4$

3) $-2 \geq k$

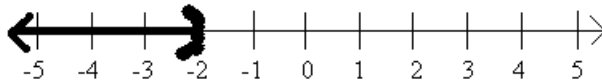
4) $1 \geq k$

5) $5 \geq x$

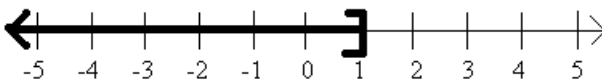
6) $-5 < x$

Write an inequality for each graph.

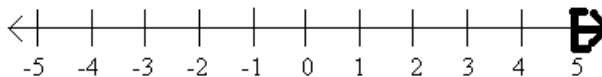
7)



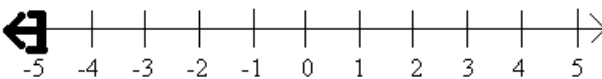
8)



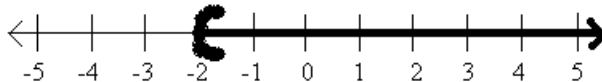
9)



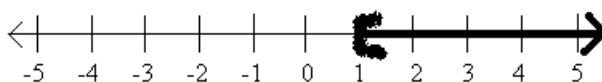
10)



11)



12)



Solve each inequality, graph each solution, and give interval notation.

13) $\frac{x}{11} \geq 10$

14) $-2 \leq \frac{n}{13}$

15) $2 + r < 3$

16) $\frac{m}{5} \leq -\frac{6}{5}$

17) $8 + \frac{n}{3} \geq 6$

18) $11 > 8 + \frac{x}{2}$

19) $2 > \frac{a-2}{5}$

20) $\frac{v-9}{-4} \leq 2$

21) $-47 \geq 8 - 5x$

22) $\frac{6+x}{12} \leq -1$

23) $-2(3+k) < -44$

24) $-7n - 10 \geq 60$

25) $18 < -2(-8+p)$

26) $5 \geq \frac{x}{5} + 1$

27) $24 \geq -6(m-6)$

28) $-8(n-5) \geq 0$

29) $-r - 5(r-6) < -18$

30) $-60 \geq -4(-6x-3)$

31) $24 + 4b < 4(1+6b)$

32) $-8(2-2n) \geq -16+n$

33) $-5v - 5 < -5(4v+1)$

34) $-36 + 6x > -8(x+2) + 4x$

35) $4 + 2(a+5) < -2(-a-4)$

36) $3(n+3) + 7(8-8n) < 5n+5+2$

37) $-(k-2) > -k-20$

38) $-(4-5p) + 3 \geq -2(8-5p)$